# SCI-FLT-FIXED v0.1 Engineering-Conformance Specification

SCI-FLT-FIXED-ENGINEERING-CONFORMANCE v0.1/draft-r0.4

**Status: implementation-blind Stage B conformance-target draft; no conformity finding**

**Scientific owner: Grant Wilson**

**Stage B date: 2026-08-31**

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Source SHA-256: 6d5d648397f7cb463ffc9117b9a45f89a9918613a67adfaff598e817b34bc471
Shared normative core SHA-256: 43c1a5f57cb72b03cfc4a99628ae4c86fcabdf0a338cac2cdc12e08774010999
Author packet manifest SHA-256: 7f2d03f182258ac9770f7dba869e9ae0b5018efdcdb93b18b299a9b9c6df1e4d
r0.2 owner directive SHA-256: 02ca63dcc8ac19e554fa333c20773ff85264336b4f622024605f6514ec1716d1
r0.3 owner directive SHA-256: c1de5ad5b9f9e5451ec2a3d9d9c18d30f7990eec721497c9d473aaeb2b043551
r0.4 owner directive SHA-256: 855d9709d3fe3b4d71b6a56a020e52ce52164a0384a2ed2d562145f8fb3b0a99
Build recipe SHA-256: 145078013207cbd78fcc778ee8f2e73dbc505663e72c8ac47c6705ae39fdf7f0
```

Implementation-blind scientific-contract draft. This artifact reports no implementation, validation, calibration, achieved-response, achieved-covariance, performance, readiness, freeze, production, or Unity result.

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Normative import: the complete `SCI-FLT-FIXED-NORMATIVE-CORE v0.1/draft-r0.4`, source SHA-256 `43c1a5f57cb72b03cfc4a99628ae4c86fcabdf0a338cac2cdc12e08774010999`, is incorporated without modification. This view adds no scientific rule. If this view and the imported core differ, the core controls.

## 1. Conformance target and evidence boundary

This specification translates the shared scientific core into observable engineering obligations without describing, inspecting, or judging an implementation. A future conformance review must bind one candidate realization to one exact core revision, parent generation, resolved operator generation, and product generation.

A claim passes only when positive evidence establishes the exact required identity and behavior. A missing, conflicting, approximate, same-name, or inferred fact yields unavailable or failed status as directed by the core. A finite array, successful process exit, or familiar label is not sufficient.

Conformance, scientific validation, calibration, performance, readiness, and production are separate evidence classes. This specification defines only a future conformance target.

## 2. Required input and plan records

A candidate realization must accept an immutable request record, effective decision, complete parent binding, and externally resolved fixed plan. The records must expose enough content to reconstruct and compare exact identities, not merely display friendly names.

The parent record must distinguish MAP observation, MAP coadd, and JINC observation roles. It must bind the complete applicable upstream identity, quantity, units, nominal-beam, WCS, support, validity, response, covariance, null, exposure, lifecycle, failure, and provenance state. JINC requires all five atomic numerical roles. Upstream unavailability must remain explicit.

The record must identify exactly one signal vector: the base MAP observation signal, base MAP coadd signal, or normalized `jinc_map`. It must retain exact JINC roles `jinc_signal_numerator`, `jinc_signed_normalization`, `jinc_quadratic_accumulator`, `jinc_map`, and `jinc_coefficient_squared_time`, while proving that only `jinc_map` becomes `FLT-SIG`. Other map-shaped roles remain separately typed facts or compositions.

The plan record must distinguish requested, effective, disabled, unavailable, and resolved states. A resolved plan must identify `FLT-FIXED-CONV` as the sole base family, bind one exact sampled convolution, and bind every coefficient, parameter, coordinate domain, grid and metric fact, normalization, support and edge rule, transfer qualification, and provenance fact before application. "Externally resolved" does not transfer FLT policy or application ownership.

The base `m`, `y`, `k_Theta(r)`, `L_Theta`, and `A_Theta,J` records must be real-valued. Each coefficient must be finite, real, unit-typed, canonically exactly represented, and content-bound. Missing, non-finite, complex, unrepresentable, or conflicting coefficients make the plan unavailable before application; a numerical-comparison tolerance may not repair them. A complex transfer `H(nu)` remains only a representation of the real operator.

## 3. Operator reconstruction record

`FLT-OPERATOR` must provide a content-bound representation sufficient to reconstruct the exact finite `L_Theta`, frozen `J_full`, and complete `A_Theta,J`, including:

- ordered input and output row domains;
- exact WCS, frame, topology, metric, shape, indexing, and pixel-area facts;
- complete sampled coefficients, coefficient units and ordering, numerical representation, and digest;
- representation-invariant `K_geom_science`, nonauthoritative `K_store`, `K_nonzero`, ordinary-method `K_req = K_nonzero`, canonical exact-zero representation, center, extent, tie, phase, subpixel convention, orientation, and handedness;
- declared normalization, DC gain, and distinct signed, absolute, squared, and geometric support summaries;
- full-footprint predicates, exact inherited zero-operator row domain, and cause vocabulary;
- qualified transfer facts or explicit unavailability; and
- immutable plan and operator generation identifiers.

The record must show that one product applies this exact sampled convolution once and makes no intermediate or reordered-composition claim. An alternate computational mechanism is comparable only after reconstructing the identical declared operator. A different coefficient representation that changes a declared coefficient, ordering, phase, row domain, or numerical operator is not scientific equivalence.

## 4. Admission and application gate

Bundle admission must evaluate `SCI-FLT-FIXED:input_bundle_admission@1` against one exact request, parent, and resolved plan. Parent-row admission must evaluate `SCI-FLT-FIXED:input_parent_row_admission@1` for every exact parent row and named FLT use. Each profile returns the typed request, applicability, eligibility, missing/conflict, action, and cause state defined by its bound profile. Neither performs payload arithmetic or authors output-row support.

Application may proceed only for an eligible resolved route. Plan state must be independent of parent amplitudes. The selector must be resolved once before convolution arithmetic from declared immutable parent identity and row membership, exact input admission, typed availability and finiteness, support, and required predicates. Inspecting an authorized required location solely to classify its declared finite state is structural screening and must not tune the plan. For every output row, FLT verifies all `K_req` locations against those exact facts. The scientific row domain must equal the passing set. Response perturbations, covariance draws, noise realizations, and NOI members must reuse that selector; a member failure produces row unavailability, not re-selection.

No extension, periodic wrap, truncation, local renormalization, inpainting, reflection, clamp, mirror, padding-based admission, edge completion, or value replacement is conforming in v0.1. Rows failing the full-footprint gate remain unavailable with causes, even if a storage payload is finite.

Dense, sparse, cropped, and zero-padded encodings of one canonical kernel must reconstruct identical scientific support, row admission, response, covariance, and product identity. Only `K_store` may differ. Ordinary convolution arithmetic must traverse exactly `K_nonzero = K_req`, not `K_geom_science`. An exact-zero offset creates no arithmetic, payload, influence, covariance, or admission dependency, and its parent payload must not be evaluated or dereferenced.

A requested nonzero convolution with empty `S_out` must retain its application evidence as `applied_no_scientific_output_support`. Publication must prescribe `not_produced` with cause `no_full_footprint_output_rows`; it must not emit an empty `FLT-SIG` or relabel the state as disabled, failed execution, identity, or zero.

## 5. Signal, response, transfer, mode, and influence records

Application must compute `FLT-SIG` from the exact resolved operator on the exact scientific row ordering. The output-unit record must derive units from parent and coefficient units and retain the originating nominal-beam and calibration lineage without introducing a filtered-beam label.

Each response record must name fixed-state linear, already realized parent- grid, parent full-procedure finite difference with FLT fixed, or FLT re- resolved procedure family. The first three must apply the identical frozen `A_Theta,J` exactly once; the fourth is outside v0.1. A full-procedure difference must bind compatible baseline and perturbed membership, availability, WCS, quantity, and support on the frozen required domain. A mismatch retains the parent state change, does not re-resolve `J_full`, and makes affected response rows unavailable. Missing basis, domain, or parent-response identity is explicitly unavailable. The exact zero operator records its local zero derivative without promoting it to an unavailable complete source response. Kernel-only, approximate, multiply applied, differently centered, differently normalized, or differently edged surrogates do not pass.

Transfer and mode records must bind the exact coordinate domain, WCS metric, transform sign and normalization, coordinate/frequency units, origin, ordering, signed/Nyquist treatment, frequency grid, response quantity, attenuation units, band geometry, phase branch, finite-grid domain, null, invariant, anisotropy, and attenuation facts that are scientifically defined. They must distinguish sampled-kernel transfer from the complete row-restricted operator. Unsupported facts remain unavailable. Influence exposes the parent-row coefficient relation; `FLT-EXPOSURE-LINEAGE` binds parent exposure identity or typed absence and confirms that neither influence nor a convolved exposure plane is physical exposure.

## 6. Covariance and NOI records

For an available compatible parent covariance, a covariance-qualified candidate must realize the exact two-sided frozen-operator propagation on `S_out`. Its record must state parent stochastic authority separately from output representation and name the conditional model, omitted terms, excluded selection/kernel/beam/WCS/calibration/model uncertainties, domain, ordering, rank, null space, and supported operations.

If an explicitly diagonal parent covariance is propagated, any complete output representation must include induced cross-row covariance. A marginal-only representation must say so. Missing cross terms may not be encoded as zero or independence. For marginal-only parent authority, a row with exactly one nonzero coefficient has the exact conditional marginal `A_ij^2 Var(m_j)`; a row mixing two or more parent variables remains unavailable or explicitly partial. The one-sparse result authorizes no cross-row covariance or independence. The exact-zero row keeps its separately typed zero parent-payload contribution. Structured or partial authority permits only proved-exact operations. Any diagonal-contribution diagnostic must avoid variance, covariance, uncertainty, and precision labels.

`FLT-NOI-COMPATIBILITY` must be immutable publication-time FLT state containing no future NOI identity. A later NOI child references FLT; an optional reverse relation is separately versioned and cannot mutate FLT. Every admitted NOI member must receive the identical frozen `A_Theta,J`; member footprint failure is unavailable. The route rejects filtering uncertainty-like products, approximate transfer, relocation, commutation, substitution, and any per-member re-resolution.

A `not_requested_at_FLT_publication` state is historical provenance only. It must not block a later independently requested compatible NOI child, whose request, applicability, eligibility, realization, generation, and failure are child-owned and cannot mutate FLT.

## 7. Atomic publication and lifecycle records

Publication policy must evaluate `SCI-FLT-FIXED:output_publication@1` against one complete publication candidate. The candidate contains all required roles from the imported core, including request-specific honest unavailable companion records. A partial candidate, placeholder, or inferred companion fails publication. The policy defines disposition and prescribed action. VAL may produce a decision artifact but performs no publication. The FLT publisher performs or declines the action and owns final realization and local validity.

Lifecycle evidence must distinguish `not_requested`, `requested`, `effective`, `disabled`, `unavailable`, `resolved`, `applied`, `applied_no_scientific_output_support`, `complete_publication_candidate`, `publication_decision`, `not_produced`, `realization_failed`, `failed`, `realized_identity`, `realized_zero`, `realized`, and `superseded`. Disabled is not produced. Identity and zero use the same candidate and decision sequence. Required failure propagates and emits no complete product.

Every product must bind immutable parent, plan, operator, output, companion, cause, lifecycle, failure, and provenance generations. Any core-defined identity change creates a new generation. A later NOI child or reverse relation is separate and cannot mutate an existing FLT bundle.

## 8. Requirement conformance matrix

The following matrix routes every stable core requirement to a future observable conformance decision. The exact normative text remains in the imported core.

- `SCI-FLT-FIXED-REQ-001`: verify package, tranche, convolution-only base, and qualified low-pass subtype identity; reject arbitrary-linear or inference- bearing identity.
- `SCI-FLT-FIXED-REQ-002`: reconstruct `A` and verify no additive term or additive-state dependency.
- `SCI-FLT-FIXED-REQ-003`: verify exactly one supported immutable parent role.
- `SCI-FLT-FIXED-REQ-004`: verify complete applicable parent fields and atomic JINC roles.
- `SCI-FLT-FIXED-REQ-005`: verify upstream unavailability remains fail-closed.
- `SCI-FLT-FIXED-REQ-006`: verify observation, coadd, and JINC successor identities and absence of FLT coaddition or inferred commutation.
- `SCI-FLT-FIXED-REQ-007`: verify all plan and selector state predates payload arithmetic, plan state is amplitude-independent, selector structural screening uses only authorized parent facts, and companions do not re-resolve the selector from their payloads.
- `SCI-FLT-FIXED-REQ-008`: compare exact WCS and grid facts and reject approximate joins or resampling.
- `SCI-FLT-FIXED-REQ-009`: reconstruct canonical coefficients, `K_geom_science`, nonauthoritative `K_store`, `K_nonzero`, `K_req`, domains, discretization, and digest; compare representation invariance.
- `SCI-FLT-FIXED-REQ-010`: verify one-time convolution construction from the exact sampled kernel over `K_nonzero = K_req`, without geometric exact-zero, intermediate, or reordered terms.
- `SCI-FLT-FIXED-REQ-011`: verify every low-pass transfer fact or mark only the qualification unavailable.
- `SCI-FLT-FIXED-REQ-012`: recompute the full-footprint row set from `K_req = K_nonzero` and predicates independently of geometric or storage zeros, including the no-dereference rule.
- `SCI-FLT-FIXED-REQ-013`: verify all excluded rows are unavailable with causes, never promoted by storage shape.
- `SCI-FLT-FIXED-REQ-014`: detect and reject every deferred edge or replacement method.
- `SCI-FLT-FIXED-REQ-015`: verify transformed-amplitude identity and exact unit derivation without amplitude-category relabeling.
- `SCI-FLT-FIXED-REQ-016`: verify nominal-beam and CAL lineage retention and absence of unsupported beam or flux claims.
- `SCI-FLT-FIXED-REQ-017`: identify the response family, compare its one-time frozen-operator composition, or verify honest unavailability.
- `SCI-FLT-FIXED-REQ-018`: verify finite-domain transfer, mode, and phase facts or explicit unavailable states without whole-chain promotion.
- `SCI-FLT-FIXED-REQ-019`: compare influence to coefficients, verify it is not exposure, and verify exact exposure-lineage state.
- `SCI-FLT-FIXED-REQ-020`: verify distinct computability, support, admission, validity, confidence, and downstream states.

- `SCI-FLT-FIXED-REQ-021`: compare covariance to exact two-sided propagation.
- `SCI-FLT-FIXED-REQ-022`: verify separate parent-authority and output- representation axes, conditional model, exclusions, and unknown cross-term handling.
- `SCI-FLT-FIXED-REQ-023`: test induced off-diagonal terms and reject marginal planes presented as complete covariance.
- `SCI-FLT-FIXED-REQ-024`: verify empirical uncertainty and significance are absent from FLT ownership.
- `SCI-FLT-FIXED-REQ-025`: compare every NOI member's full operator state to the signal state.
- `SCI-FLT-FIXED-REQ-026`: detect and reject per-member selection or re-resolution.
- `SCI-FLT-FIXED-REQ-027`: verify every atomic role, exposure lineage, allowed honest unavailable record, and immutable NOI-compatibility role without a future NOI identity or completion dependency.
- `SCI-FLT-FIXED-REQ-028`: verify the complete candidate-before-publication lifecycle vocabulary, causes, failure, and generation bindings.
- `SCI-FLT-FIXED-REQ-029`: test distinct disabled, identity, and zero outcomes.
- `SCI-FLT-FIXED-REQ-030`: mutate each identity-defining fact in isolation and verify a new immutable generation; verify NOI nonmutation.
- `SCI-FLT-FIXED-REQ-031`: remove or conflict each required fact and verify fail-closed behavior without fallback.
- `SCI-FLT-FIXED-REQ-032`: evaluate distinct bundle and exact parent-row admission request, applicability, eligibility, unavailable, exclusion, and cause branches while FLT constructs output support.
- `SCI-FLT-FIXED-REQ-033`: evaluate the complete publication candidate, disposition, prescribed action, request-specific honest-unavailable, partial, disabled, identity, zero, and failed branches; verify the publisher alone performs the action.
- `SCI-FLT-FIXED-REQ-034`: verify VAL only binds or evaluates an immutable approved policy and authors no FLT fact, arithmetic, publication, or realization.
- `SCI-FLT-FIXED-REQ-035`: verify no generic downstream admission follows from product availability.
- `SCI-FLT-FIXED-REQ-036`: verify every excluded family and every Stage B nonclaim remains outside the product claim set.
- `SCI-FLT-FIXED-REQ-037`: freeze `J_full` once and verify identical selector reuse, plan amplitude-independence, exact parent-fact structural screening, and explicit exclusion of unsupplied selection uncertainty.
- `SCI-FLT-FIXED-REQ-038`: verify `K_geom_science`, `K_store`, `K_nonzero`, ordinary `K_req = K_nonzero`, exact-zero, identity, and declared zero- operator support semantics.
- `SCI-FLT-FIXED-REQ-039`: evaluate all three typed policy objects and each request-specific companion qualification.
- `SCI-FLT-FIXED-REQ-040`: distinguish every response family and prevent local zero derivative from becoming an unsupported complete response claim.
- `SCI-FLT-FIXED-REQ-041`: verify exactly one sampled convolution application and absence of composition, collapse, or reordering claims.
- `SCI-FLT-FIXED-REQ-042`: verify preregistered numerical comparison fields, frozen bounds, independent oracle, and nonclaim boundary.
- `SCI-FLT-FIXED-REQ-043`: verify exact inherited exposure lineage and reject influence or convolved exposure as physical exposure.
- `SCI-FLT-FIXED-REQ-044`: verify external timing while retaining exact FLT-owned plan, selector, application, and publication ownership.

- SCI-FLT-FIXED-REQ-045: verify only the exact parent signal role becomes FLT-SIG and all JINC diagnostics, support, validity, exposure, response, and covariance retain their typed routes.
- SCI-FLT-FIXED-REQ-046: exercise every covariance-authority row and reject unproved marginal, structured, partial, or unavailable strengthening while admitting only the exact one-sparse conditional marginal edge case.
- SCI-FLT-FIXED-REQ-047: inspect immutable FLT-NOI-COMPATIBILITY content and prove a publication-time not-requested state does not block a later independent child and that children and reverse relations do not mutate FLT.
- SCI-FLT-FIXED-REQ-048: verify bundle/parent-row profile decisions, FLT selector construction, VAL decision-artifact boundary, and publisher action.
- SCI-FLT-FIXED-REQ-049: compare every bound low-pass transform convention and reject a complete but different convention.
- SCI-FLT-FIXED-REQ-050: compare full-procedure parent domains and verify mismatch unavailability without selector re-resolution.
- SCI-FLT-FIXED-REQ-051: reproduce every authority-manifest path, byte count, SHA-256, role, compatibility or supersession state, and generated-view relation or route the dependent claim to typed unavailability.
- SCI-FLT-FIXED-REQ-052: verify the real scalar field and reject every missing, non-finite, complex, unrepresentable, or conflicting coefficient before application without tolerance repair.
- SCI-FLT-FIXED-REQ-053: exercise empty S_out for a requested nonzero convolution and verify the exact application state, publication action, cause, retained evidence, and forbidden relabelings.

## 9. Falsifiable prediction suite

Each future prediction evaluation must bind exact inputs, operator state, expected row domain, expected companions, expected lifecycle, comparison semantics, and causes before execution. The normative expected outcomes are in the imported core.

- SCI-FLT-FIXED-PRED-001: identity operator and identity lifecycle.
- SCI-FLT-FIXED-PRED-002: zero operator, inherited parent-support domain, local zero derivative and covariance, honest unavailable complete response, and zero lifecycle.
- SCI-FLT-FIXED-PRED-003: scalar input linearity on one frozen selector.
- SCI-FLT-FIXED-PRED-004: constant input under exact DC gain and every authorized normalization.
- SCI-FLT-FIXED-PRED-005: impulse, center, orientation, phase, indexing, and exact-nonzero arithmetic support.
- SCI-FLT-FIXED-PRED-006: exact separately typed response-family composition with one frozen operator application.
- SCI-FLT-FIXED-PRED-007: signed coefficients and distinct support summaries.
- SCI-FLT-FIXED-PRED-008: zero-sum constant-mode null without claim promotion.
- SCI-FLT-FIXED-PRED-009: full-footprint inclusion, dependent-row removal, storage invariance, and exact-zero no-dereference behavior.
- SCI-FLT-FIXED-PRED-010: rejection of every deferred edge and replacement method, including support-conditioned renormalization.
- SCI-FLT-FIXED-PRED-011: missing, unavailable, non-admitted, and non-finite required parent locations.
- SCI-FLT-FIXED-PRED-012: exact complete covariance on ordered S_out.

- `SCI-FLT-FIXED-PRED-013`: induced cross-row covariance conditional on an explicit independent-diagonal parent model.
- `SCI-FLT-FIXED-PRED-014`: unavailable parent response or covariance and rejection of surrogates.
- `SCI-FLT-FIXED-PRED-015`: exact WCS and grid mismatch failure.
- `SCI-FLT-FIXED-PRED-016`: distinct observation, coadd, and JINC identities with no FLT coadd product.
- `SCI-FLT-FIXED-PRED-017`: exact frozen-selector NOI member parity and member-level unavailable footprint state.
- `SCI-FLT-FIXED-PRED-018`: per-member re-resolution rejection.
- `SCI-FLT-FIXED-PRED-019`: disabled, identity, zero, and failure lifecycle distinctions.
- `SCI-FLT-FIXED-PRED-020`: upstream unavailable MAP and JINC route.
- `SCI-FLT-FIXED-PRED-021`: low-pass qualification completeness.
- `SCI-FLT-FIXED-PRED-022`: exact-zero canonical support and storage invariance.
- `SCI-FLT-FIXED-PRED-023`: zero-operator inherited row-support domain.
- `SCI-FLT-FIXED-PRED-024`: independent exact sampled low-pass transfer under the identical complete transform convention.
- `SCI-FLT-FIXED-PRED-025`: rejection of every non-signal parent role as ordinary-method `FLT-SIG`.
- `SCI-FLT-FIXED-PRED-026`: filtered-variance sensitivity to parent cross terms with identical marginal diagonals plus one-sparse, two-sparse, and exact-zero marginal-only fixtures.
- `SCI-FLT-FIXED-PRED-027`: later NOI child and reverse-relation nonmutation of the immutable FLT bundle, including a late request after publication-time not-requested provenance.
- `SCI-FLT-FIXED-PRED-028`: full-procedure response domain incompatibility and frozen-selector preservation.
- `SCI-FLT-FIXED-PRED-029`: empty scientific output support, retained application evidence, exact not-produced cause, and forbidden relabelings.
- `SCI-FLT-FIXED-PRED-030`: invalid-coefficient rejection before application with no numerical-tolerance repair.

## 10. Numerical-comparison evidence record

A future candidate comparison must bind one immutable policy before results are observed. The evidence record must identify its independent exact or high- precision oracle; absolute, relative, and near-zero rules; signed-cancellation and zero-sum cases; conditioning and operation-count dependence; covariance comparison; sequential and parallel agreement; overflow, underflow, and non- finite handling; simultaneous row decisions; lifecycle; and provenance.

The record must prove that bounds were not changed after a failure and that scientific coefficients and identity remained exact. Passing such a policy would be future conformance evidence only. This specification supplies no candidate result, numerical-adequacy finding, validation, or performance claim.

## 11. Traceability, identity, and reproducibility record

`TRACEABILITY.json` maps every requirement and prediction identifier to its normative-core section, scientist-rationale section, engineering-conformance section, exact admitted Stage A object sections, and exact r0.2, r0.3, and r0.4 owner-directive sections where a revision changes semantics. The durable verifier must reject missing, duplicate, extra, malformed, or unadmitted trace entries.

`BUILD_BINDING.json` records the packet manifest binding, all 17 admitted object hashes and byte counts, exact r0.2, r0.3, and r0.4 owner-directive bindings, every Stage B source and build-tool hash, deterministic build environment, output PDF identities, page counts, byte sizes, and SHA-256 digests. The source-closure, owner-decision parity, core/view parity, semantic-change, profile, rebuild, and visual-QA records are required. A clean separate rebuild must reproduce the three PDF digests exactly.

`AUTHORITY_MANIFEST.json` is the superseding proposed-freeze inventory. It binds the Stage A packet, all three directives, boundaries, policy and numerical drafts, traceability, build binding, closure/parity/change/profile/rebuild/ visual reports, all sources, tools, and PDFs by exact path, bytes, SHA-256, role, compatibility or supersession state, and generated-view relation. Its own external digest provides the single entry point without a self-hash.

PDF text must expose the document identity, exact source digest, exact shared core digest, exact packet-manifest digest, and exact builder digest. PDF page rendering and visual QA are required in addition to content and hash checks.

## 12. Nonclaims

This conformance specification does not identify, inspect, modify, or judge an implementation. It records no implementation conformity, algorithm change, validation result, calibration state, achieved response or covariance, numerical adequacy, performance, readiness, scientific freeze, production status, or Unity activity.